

PowerTools for Autodesk Fusion

PowerTools is an Autodesk Fusion all-in-one Add-In that offers a suite of productivity tools for assembly management, document, and data tools, part-modeling, exports, related-data templates, and document sharing. It is deployed as a single installable Add-In. It adds commands to the Design workspace toolbar, the Quick Access Toolbar (QAT), and the Drawing workspace that reduce the steps required for design, data, and collaboration tasks.

The add-in includes a dedicated preferences command which allows customization of commands, as well as enable/disable each command. The tools are separated into functional categories:

- **Assembly Management**
- **Document and Data Tools**
- **Part Modeling Helpers**
- **Exports**
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- **Document Sharing**

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Prerequisites

Before you install and run this add-in, confirm that you have the following:

- **Autodesk Fusion** (any current subscription tier) with Python add-in support enabled
- **Windows 11** or **macOS**
- An **Autodesk Team Hub** (required for commands that access cloud document references, related data, and sharing)

Installation

1. Download or clone this repository to your local machine.
2. In Autodesk Fusion, open the **Add-Ins** dialog by selecting **Utilities > Add-Ins**, or press **Shift+S**.
3. On the **Add-Ins** tab, click the green **+** icon.
4. Navigate to the folder where you placed the add-in files and select the **PowerTools** folder.
5. Click **Open**.
6. Select **PowerTools** in the list, then click **Run**.

To have the add-in load automatically each time Fusion starts, select **Run on Startup** before clicking **Run**.

Commands

Most commands appear in **Design › Tools › Power Tools**. Some appear in other locations as noted below. Click a command for its detailed guide.

Assembly

Command	Location	Description
Assembly Builder	Design › Tools › Power Tools	Build and manage multi-component assemblies from a guided palette. Allows fast creation of assembly structures at the start of the design.
Assembly Palette	Design › Solid	Quick-start palette for populating a new assembly: create components in place, or insert from open and recent documents. asses assembly builder, global parameters a fastener shortcuts.
Insert Step	Design › Solid	Insert a STEP file as a referenced component.
Assembly Statistics	Design › Tools › Power Tools	Report component, occurrence, and reference counts for the active design.
Get and Update	Design › Tools › Power Tools	Fetch and update referenced documents in one step. Often more reliable than Fusion's own update process.
Bottom-Up Update	Design › Tools › Power Tools	Propagate changes from sub-components up through the assembly.
Externalize	Design › Tools › Power Tools	Externalize internal components into documents replacing them in the active assembly keeping positions intact. Great for converting imported assemblies to distributed designs.
Global Parameters	Design › Tools › Power Tools	View and manage project-wide global parameters.
Link Global Parameters	Design › Tools › Power Tools	Link the active design to project global parameters.
Refresh Global Parameters Cache	File › PowerTools Settings	Rebuild the cached global-parameters folder and document index.
Infer Constraints	Design › Tools › Power Tools	Infer assembly joints/constraints from component geometry. Great for creating joints/constraints on imported assemblies.
Component Warning	File › PowerTools Settings	Toggle warnings about creating features outside a component.
Change Cycle Color	Design › Right-click menu	Set the per-component color used by Fusion's Color Cycling Toggle on selected components. Better than selecting fusions random colors.
Reference Manager	Design › Tools › Power Tools	Inspect and manage all external references of the active design.
Document References	Design › Tools › Power Tools	List all documents related to the active design. Shows, parents, drawings, Top level (root) assemblies, fasteners, child components, and any Related Data .
Document Refresh	Design › Tools › Power Tools	Refresh out-of-date active document to latest version on cloud. Great when working in a team and others have update a document you have open.

Document Tools

Command	Location	Description
Document Information	Design › Tools › Power Tools	Display cloud data identifiers and metadata for the active document.
Document History	Quick Access Toolbar	Show the active document's version history as a palette of day rows: a track per author, saves on a clock axis, and the elapsed time between days.
Version Diff	Design › Tools › Power Tools	Compare versions of the active document.
Assign Part Numbers	Design › Tools › Power Tools	Assign part numbers across the active design.

Command	Location	Description
Assign Drawing Number	Drawing › Power Tools	Assign a drawing number inside the Drawing workspace.
Sync Item to Part Number	Design › Manage › Power Tools	Copy the Fusion Manage Item Number into the Part Number (Manage Extension).
Default Folders	Design › Tools › Power Tools	Create default project folders in a new project using one of several configured folder schemes.
Favorites	Design › Tools › Power Tools	Manage favorite documents and folders. Also allows quickly navigating to favorites in the data panel.
Open Recent	File › Open Recent	Reopen a recently used document from a File-menu flyout, with location and thumbnail tooltips.
Match Units	Design › Inspect	Compare the active document's length and mass units with your Fusion default units and change the document to match in one click. The button's icon reports whether they agree. Optional prompts catch a mismatch when a document opens, and when you switch to the Manufacture workspace, where Fusion keeps a separate unit system that silently changes your feeds and speeds.
Show In Location	File › PowerTools Settings	Open the active document's location in the data panel. Can be set to open on document open and document tab switch.
Close All Documents	File	Close every open document, saving or discarding unsaved changes as a group.
Toggle Data Pane	Navigation bar	Toggle the visibility of the data panel.
Recovery Save	Design › Tools › Power Tools	Direct access to creating a recovery auto-save for the active document.

Exports

Command	Location	Description
Export BOM	Design › Tools › Power Tools	Export the bill of materials to CSV.
Export Mermaid	Design › Tools › Power Tools	Export the assembly structure as a Mermaid diagram.
Export SysML	File	Export the assembly as a 4+1 architecture document with a SysML physical view.

Part Modeling

Command	Location	Description
Sketch Repair	Sketch › Modify	Repair common sketch profile issues automatically.
Round Sketch Dimensions	Sketch › Modify	Round the active sketch's dimensions to an adjustable increment. Especially useful for updating a sketch after auto-constrain.
Under-Constrained Sketch	Sketch	Highlight under-constrained sketch geometry.
Radial Hole Circle	Sketch	Add center points for a radial pattern of holes.
Mirror Derive	Design › Solid	Create a mirrored derived component.
Hide Objects	Design › Tools	Quickly hide objects from a list of options.
Timeline Compute Times	Design › Solid	Measure per-feature timeline compute times and display a relative report.
Measure Path	Design › Inspect	Total the length of a chain of edges and sketch curves.
Flatten Surface	Design › Tools	Flatten curved faces to a sketch pattern, with a stretch/gather strain map.

Animation

Command	Location	Description
Save Named View	Animation › Power Tools	Save the Animation viewport camera as a named view on the design, named from the active storyboard. Very handy for saving the view in which you create an exploded view in the animation workspace.

Related Data

Command	Location	Description
Create Related Data	Design › Solid	Create a new document referencing the active document from a folder in team of pre-created start documents.
Select Related Data Folder	File › PowerTools Settings	Configure the hub/project/folder used for related data start documents.

Team Add-ins

Share add-ins across a team through one hub folder — **Assets / Shared Addins** in the active hub. Drop an add-in **.zip** in; everyone else picks it up shortly after Fusion starts, with no restart and nothing to publish. Dormant until the folder exists, and silent unless something changed.

Command	Location	Description
Team Add-ins	Design › Tools › Power Tools	Check the shared folder now and show what the last check did.
Set Up Shared Add-ins Folder	File › PowerTools Preferences	Find or create the shared folder in the hub's Assets project.

Tools

Command	Location	Description
Scripts and Add-ins	QAT › File menu	Open Fusion's built-in Scripts and Add-Ins Manager from the File menu. When using home tab, this allows access to the scripts and Add-Ins Manager when no document is open.

Share

Command	Location	Description
Get a Share Link	QAT › Share Menu	Create and copy a public share link for the active document.
Change Share Settings	QAT › Share Menu	Manage download and password protection for the share link.
Get Open on Desktop Link	QAT › Share Menu	Get a link that opens the document in the Fusion desktop client.
Get Open in Team Link	QAT › Share Menu	Get a link that opens the document in Autodesk Fusion Team web client.
Invite to Project	QAT › Share Menu	Invite a collaborator to the active project.
Document Project Members	QAT › Share Menu	List the members of the active project.

Dev-Notes

For developer-oriented documentation — system context, C4 diagrams, the add-in lifecycle, the shared-access-point model, and the command-module pattern — see [docs/arch/architecture.md](#).

For local development setup, tooling, and how to debug the add-in in VS Code or Zed, see the [developer guide](#).

AI coding agents start at [AGENTS.md](#), which links the [codebase map](#) and the [lessons ledger](#).

Support

This add-in is developed and maintained by IMA LLC.

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The three Autodesk-sample-derived modules in `lib/ptAddInUtils` — `general_utils.py`, `event_utils.py`, and `attributes_utils.py` — remain under Autodesk, Inc.'s original, permissive license terms (see each file's header), which require that their copyright notice be retained in all copies. The proprietary terms above do not apply to those Autodesk-derived portions.

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